

hw 0518

14.4

⑩ H_0 : 結婚率和教育程度無多
 H_a : " 有少

$$E_{Yes} = \frac{300 \times 448}{900} = 166$$

$$E_{No} = \frac{300 \times 402}{900} = 134$$

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

$$= \frac{(189 - 166)^2}{166} + \dots + \frac{(151 - 134)^2}{134} = 10.061$$

$$df = (3-1)(2-1) = 2$$

$$\alpha = 0.01 \quad \chi^2_{2,0.01} = 9.210$$

$\Rightarrow$ Reject H_0

⑫

	A	D	S	
1	8	20	1	29 $\rightarrow E_1 = \frac{29 \times 40}{120} = 9.667$
2	16	8	9	33 $\rightarrow E_2 = \frac{33 \times 40}{120} = 11$
3	10	10	14	34 $\rightarrow E_3 = \frac{34 \times 40}{120} = 11.33$
4	6	2	16	24 $\rightarrow E_4 = \frac{24 \times 40}{120} = 8$
	40	40	40	120

H_0 : 三種住宅的家庭人數分佈相同

H_a : " 不同

$$\chi^2 = \sum \frac{(O - E)^2}{E} = \frac{(8 - 9.667)^2}{9.667} + \dots + \frac{(16 - 8)^2}{8} = 36.499$$

$$df = (4-1)(3-1) = 6$$

$$\chi^2_{0.01,6} = 16.812 < 36.499$$

$\Rightarrow$ Reject H_0

15.1

(12)

Above	4.8	5.2	5.0	4.9	5.1	34
Below	5.0	4.7	4.9	4.8	4.9	21

a. H_0 : above, below 分佈相同 H_a : below 的 dissolved O_2 較低

$$T_1 = 34, T_1^* = 5(5+5+1) - 34 = 21, T = \min(34, 21) = 21$$

$$\alpha = 0.05, n_1 = 5, n_2 = 5 \Rightarrow \text{查表得臨界值} = 19$$

$$T = 21 > 19 \Rightarrow \text{Non-Reject } H_0$$

b. $H_0: \mu_1 - \mu_2 \leq 0$

$\bar{X}_A = 5, s_A^2 = 0.025$

$H_a: \mu_1 - \mu_2 > 0$

$\bar{X}_B = 4.86, s_B^2 = 0.013$

$s_p^2 = \frac{4 \times 0.025 + 4 \times 0.013}{5+5-2}$

$$t = \frac{\bar{X}_A - \bar{X}_B}{\sqrt{s_p^2(\frac{1}{n_1} + \frac{1}{n_2})}} = \frac{5 - 4.86}{\sqrt{0.019(\frac{1}{5} + \frac{1}{5})}} = 0.019$$

$$= 1.606 < t_{8, 0.05} = 1.860$$

 $\Rightarrow \text{Non-Reject } H_0$

(13)

 H_0 : 兩母體的眼 movement rates 分佈相同 H_a :

分佈不同

$$T_1 = 126, T_1^* = 9(9+9+1) - 126 = 45, T = \min(126, 45) = 45$$

$$n_1 = 9, n_2 = 9, \alpha = 0.05, \text{臨界值} = 62$$

$$T = 45 < 62 \Rightarrow \text{Reject } H_0$$

在 $\alpha = 0.05$ 下, 有充足 evidence 說分佈不同

15.2

(14)

$$\begin{array}{ccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 1 & & & & & & \\ 2 & & & & & & \end{array} \Rightarrow X=6$$

 H_0 : 兩母體分佈相同 H_a : 不同

$\alpha \leq 0.1, X \sim \text{Bin}(7, 0.5)$

$$P(X=0) = \binom{7}{0} 0.5^7 = 0.0078$$

$$P(X=7) = \binom{7}{7} 0.5^7 = 0.0078$$

$$P(X \leq 1) + P(X \geq 6) = 2 \times (0.0078 + 0.0078) = 0.0264 < 0.1$$

$$P(X=0) + P(X=7) = 2 \times 0.0078 = 0.0156 < 0.1$$

$$\Rightarrow \text{R.R. } X=0 \text{ or } X=7 \Rightarrow X=6 \text{ 不落入 R.R.}$$

 $\Rightarrow \text{Non-reject } H_0$

⑪

$\mu = 11$ $\sigma = 9$

a. $H_0: p = 0.5$ $H_a: p \neq 0.5$

$X \sim \text{Bin}(20, 0.5)$

$P(X \leq 4) = 0.0059$, $P(X \leq 5) = 0.0207$, $P(X \leq 6) = 0.0597$

$\downarrow$
接近且小於 0.025

$\Rightarrow R.R. = X \leq 5$ or $X \geq 15$, $X = 11$ 不落入 R.R.

$\Rightarrow$ Non-reject H_0

b. $Z = \frac{X - \mu}{\sigma} = \frac{X - np}{\sqrt{np(1-p)}} = \frac{11 - 20 \times 0.5}{\sqrt{20 \times 0.5 \times 0.5}} = 0.447$

$Z < Z_{0.05} = 1.96$

$\Rightarrow$ Non-reject H_0

c. (a) (b) 結論相同

$\Rightarrow$ No sufficient evidence 顯示兩位 cooks 中有一位給出更高的評分

⑬

a. $\oplus = 2$, $\ominus = 8 \Rightarrow X = 2$, $n = 10$

$H_0: p = 0.5$, $H_a: p \neq 0.5$

$X \sim \text{Bin}(10, 0.5)$

$P(X = 0) = \binom{10}{0} 0.5^{10} = 0.0010$

$P(X = 1) = \binom{10}{1} 0.5^{10} = 0.0098$

$P(X = 2) = \binom{10}{2} 0.5^{10} = 0.0439$

$\Rightarrow \times (0.0010 + 0.0098) = 0.0216$

$\Rightarrow R.R. = X \leq 1$ or $X \geq 9$

$\Rightarrow X = 2$ 不落入 R.R.

$\Rightarrow$ Non-reject H_0

b. 資料非連續變態之數，
自家醫院樣本數差異極大，
導致 σ^2 不同